

Session 7:

What is Generative AI?

You've trained AI to recognize cats and dogs. You've learned how machines learn from data. Now it's time to explore the type of AI that's been making headlines - AI that doesn't just analyze, but **CREATES**. Get ready to meet the artists, writers, and musicians of the AI world!



Objectives

By the end of this session, you will be able to:

01

Define

Define Generative AI and explain how it differs from Traditional (Discriminative) AI

02

Identify

Identify whether a task requires Generative or Discriminative AI

03

Explain

Explain in simple terms how generative AI creates new content

04

List

List the main types of generative AI (text, image, audio)

05

Recognize

Recognize generative AI tools you already use in daily life

06

Use

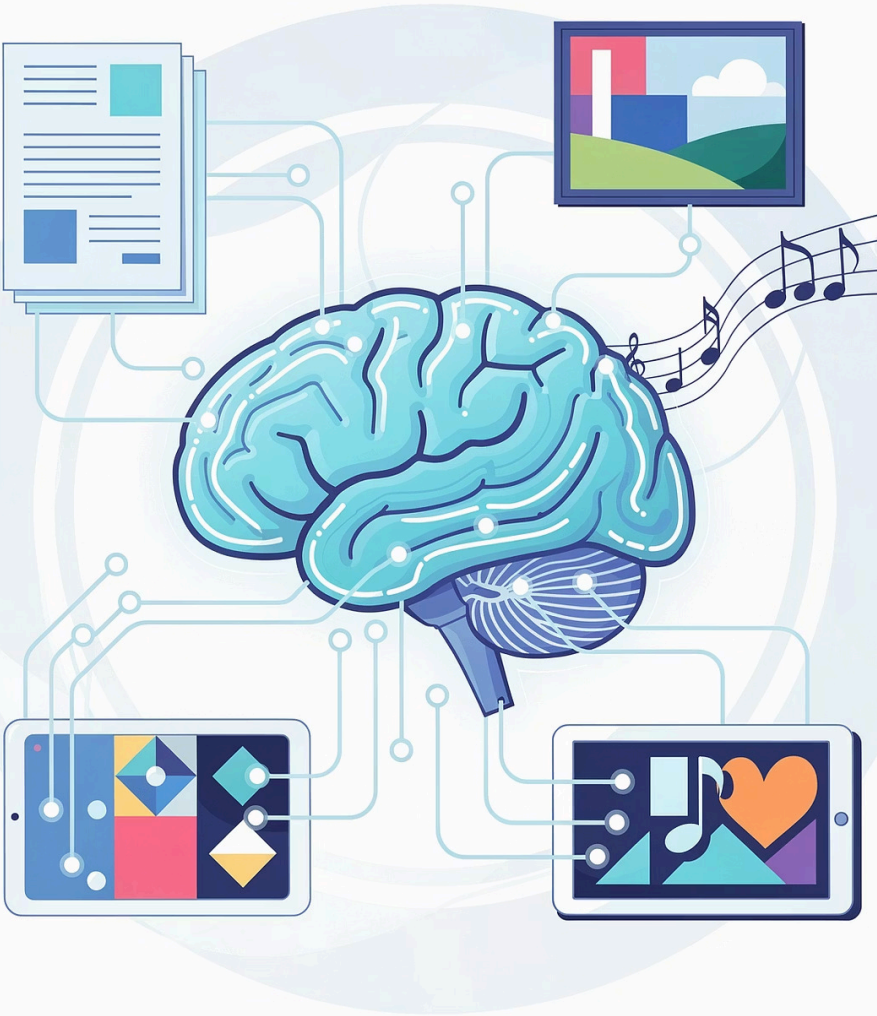
Use prompts to bring a creative idea to life with AI

Imagine It's Your Friend's Birthday

Your best friend's birthday is next week. You want to create something unique and personal - something that shows how much you care. But what if you're not sure what to make? **What if you could use AI to help?**

Idea	What Would You Need?
A funny birthday poem	A way to generate text
A custom birthday card with their favorite animal	A way to generate images
A silly birthday song about inside jokes	A way to generate audio

Today, you'll learn how AI can do ALL of these - and you'll create your own birthday artifact!



What is Generative AI?

Generative AI is a type of artificial intelligence that **creates new content** - text, images, music based on patterns it learned from training data.

Key characteristics:

Characteristic	What It Means	Example
Creates	Produces something new	A poem, picture, or song
Learns patterns	Trained on millions of examples	Studied thousands of poems, photos, songs to learn what they look/sound like
Remixes	Combines patterns in new ways	Mixes styles and ideas it has seen before
Responds to prompts	You guide what it creates	Your instructions shape the final result

Generative AI vs Traditional AI (Discriminative)

	Generative AI	Traditional (Discriminative) AI
Analogy	 Artist	 Detective
What it does	CREATES new content	ANALYZES or CLASSIFIES existing content
Question it answers	"What can I make?"	"What is this?"
Output	Original text, images, audio	Labels, categories, predictions
Birthday example	Write a poem, draw a card, compose a song	Sort party photos, check if comments are nice, count people in photos
You've used	Google Storybook, Gemini	Teachable Machine (S3, S6)



Key Insight: They're different tools for different jobs

Think of It This Way

Imagine you have two AI assistants helping with the birthday party:

Generative AI

- "Write a funny poem for my friend who loves pizza"
- "Draw a birthday card with a unicorn wearing a party hat"
- "Create a birthday song about our inside jokes"

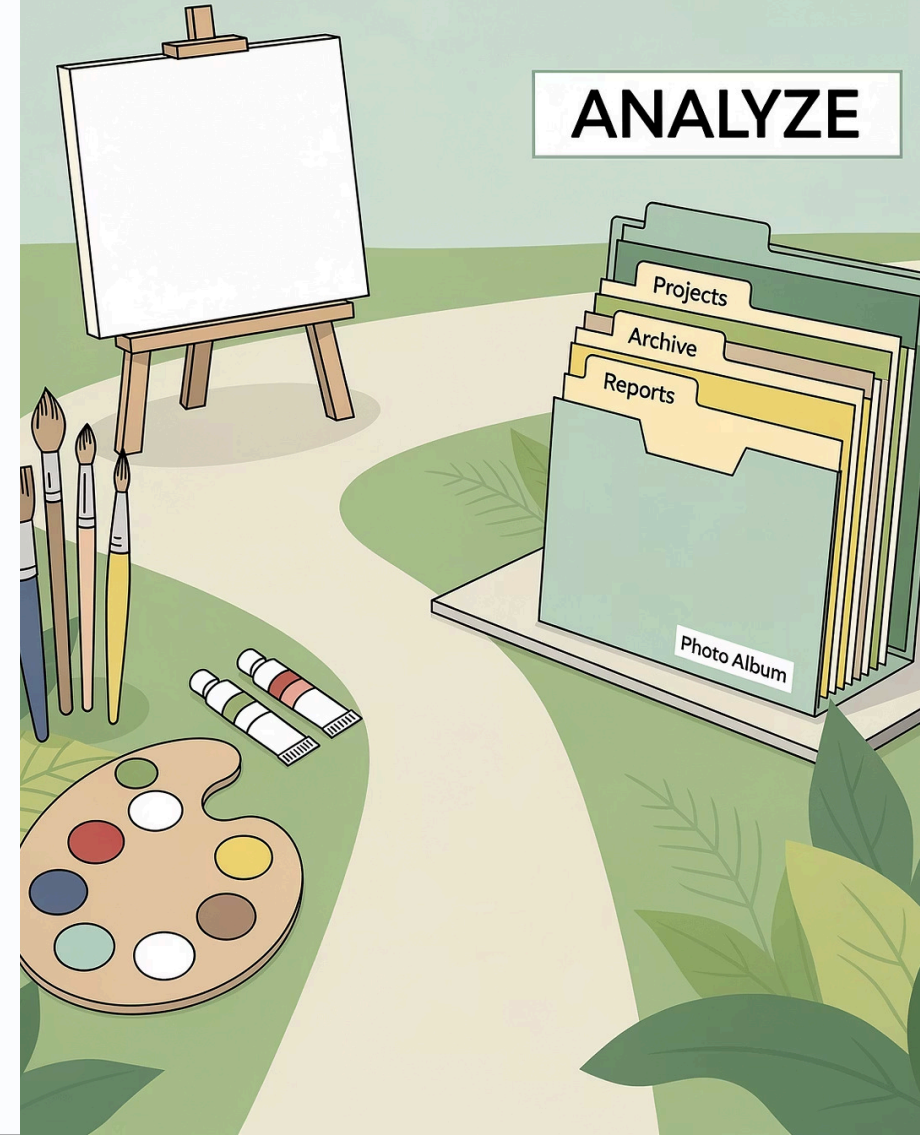
Discriminative AI

- "Sort all these party photos by who's in them"
- "Check if any comments on the party video are mean or nice"
- "Count how many people smiled in each photo"

 One **creates**. One **analyzes**. Both are useful for different tasks!

CREATE

ANALYZE



Mini Activity 1: Generative or Discriminative?

For each birthday-related task, decide: Would you use **Generative AI** (create) or **Discriminative AI** (analyze/classify)?

#	Task	Generative or Discriminative?
1	Write a funny birthday poem for your friend	
2	Sort all the party photos by who's in them	
3	Draw a birthday card with a dragon and cake	
4	Check if birthday comments on social media are positive or negative	
5	Compose a birthday song about your friend's favorite hobby	
6	Count how many people are in each party photo	
7	Create a birthday invitation text with all the details	
8	Predict how many gifts your friend might receive	
9	Make a birthday-themed coloring page for younger kids	
10	Identify whose voice is in a birthday video message	

Mini Activity 1: Answers

#	Task	Answer	Why?
1	Write a birthday poem	Generative	Creating new text
2	Sort party photos	Discriminative	Classifying/analyzing
3	Draw birthday card	Generative	Creating new image
4	Check comment sentiment	Discriminative	Classifying (positive/negative)
5	Compose birthday song	Generative	Creating new audio
6	Count people in photos	Discriminative	Analyzing/measuring
7	Create invitation text	Generative	Creating new text
8	Predict gift count	Discriminative	Predicting (Regression from S4)
9	Make coloring page	Generative	Creating new image
10	Identify voice	Discriminative	Classifying/recognizing

How Does Generative AI Work?



STEP 1: TRAINING

AI studies MILLIONS of examples



STEP 2: PATTERNS

AI learns what things
look/sound/read like



STEP 3: GENERATION

AI remixes patterns to create
SOMETHING NEW

① If the task asks you to **CREATE** something new → Generative.

If it asks you to **ANALYZE** or **DECIDE** about something existing → Discriminative.



The Remix Artist Analogy

Step 1 (Training)

They listen to pop, rock, jazz, classical - everything! They study how songs are built.

Step 2 (Patterns)

They learn patterns - how melodies work, what chords go together, how rhythms feel, what lyrics rhyme.

Step 3 (Generation)

They create a brand NEW song that sounds original - but it's built from patterns they've learned. It's a remix of everything they've heard!



Generative AI does exactly this but with **text, images, audio, etc!**

Key Point: AI Doesn't "Understand"

This is **SUPER important** - one of the most important ideas in the whole course!

Common Mistake	Reality
"AI understands what it's creating"	NO - It finds patterns, but doesn't understand meaning
"AI has original ideas"	NO - It remixes patterns it has seen
"AI knows what words mean"	NO - It knows which words usually go together

Example - When AI writes a birthday poem:

- It doesn't know what a birthday *is*
- It doesn't know what friendship *means*
- It doesn't know why birthdays are *special*
- It just knows that in millions of texts, words like "birthday," "happy," "friend," and "cake" often appear together in certain patterns

📄 Generative AI is a **PATTERN REMIXER**, not a **THINKER**.

Types of Generative AI



TEXT

- Stories, Poems, Essays
- Invitations, Jokes, Messages
- Song lyrics

*ChatGPT, Gemini, Google
Storybook*



IMAGE

- Art, Photos, Designs
- Cards, Coloring pages
- Posters, Avatars

*Gemini, Gamma, DALL-E,
Midjourney*



AUDIO

- Songs, Voice messages
- Sound effects, Birthday music
- Raps, Podcasts, Voice cloning

Suno.com, ElevenLabs

What Training Looks Like



Text

Trained on: Books, articles, websites, social media

How much? Billions of pages!



Images

Trained on: Photos, drawings, paintings, designs

How much? Millions of images!



Audio

Trained on: Songs, speeches, podcasts, sound effects

How much? Millions of hours!



Remember from Session 5: Garbage In = Garbage Out. The quality of what AI creates depends entirely on the quality of what it learned from. If the training data has biases or mistakes, the AI will reproduce them!

Example: If an AI is trained only on sad poems, it won't write happy birthday poems very well.

Mini Activity : Plan a Birthday Artifact

Your mission: You're going to create something special for a friend's birthday using AI. First, let's plan!

Step 1: Choose what you want to create (pick ONE):

Type	Birthday Artifact Ideas
TEXT	A funny birthday poem · A heartfelt birthday message · A creative invitation · A birthday-themed joke collection · A short birthday story featuring your friend
IMAGE	A custom birthday card · A coloring page · A birthday poster · A personalized avatar for your friend · A digital gift tag
AUDIO	A silly birthday song · A birthday rap · A voice message "from" a favorite character · Birthday sound effects · A personalized birthday jingle



Mini Activity : Plan Your Prompt

01

Describe your idea

Write your idea in 1-2 sentences.

02

Choose your type

What TYPE of GenAI will you need? (circle one): **TEXT** | **IMAGE** | **AUDIO**

03

Draft your prompt

Write a draft prompt here. Be specific! Include: **What you want** (poem, card, song) · **Style** (funny, heartfelt, silly, cute, formal) · **Details about your friend** (name, age, hobbies, favorite things, inside jokes) · **Format** (how long, what it should include)



We'll bring this to life in our hands-on activity!

Generative AI in Your Daily Life

You might have already used Generative AI!

TikTok

AI-generated filters and effects that create new looks for your videos

Instagram

AI-created backgrounds in stories - just type what you want and it appears!

Snapchat

"My AI" chatbot that chats with you, AI lenses that transform your face

Google Search

AI overviews that summarize information at the top of results

YouTube

AI-generated video summaries and automatic captions

Spotify

AI-generated playlists and personalized recommendations

📌 The point: Generative AI is **EVERYWHERE** in the apps you use daily. It's not a futuristic concept - it's here now, helping you create, learn, and have fun!

The Magic Word : Prompt

Prompt: The instruction or question you give to an AI to tell it what to create. *Example: "Write a funny birthday poem for my friend Alex who loves gaming"*

- 📄 Your prompt shapes everything the AI creates! The more specific and detailed your prompt, the better the result. We'll explore this more in Sessions 11-13.

Birthday-themed prompt examples:

What You Want	Your Prompt
A birthday poem	"Write a funny birthday poem for my friend Alex who loves gaming. Make it 8 lines and include an inside joke about losing at Mario Kart. Use rhymes."
A birthday card	"Create an image of a unicorn wearing a party hat, holding balloons, at a birthday party with cake. Style: cute, colorful digital art with bright colors."
A birthday song	"Compose a 30-second happy birthday song in the style of pop music. Include lyrics about friendship, cake, and having fun. Make it upbeat and catchy."

Hands-On Activity: GenAI Playground

Bring Your Birthday Artifact to Life!

Your mission: Use ChatGPT and Gemini to create the birthday artifact you planned in Mini Activity 2!

Step 1: Open both tools

Tool	Where to Find It
ChatGPT	chat.openai.com
Gemini	gemini.google.com

Step 2: Based on your artifact type, use these :

If you chose...	Use...
TEXT	ChatGPT or Gemini (try both!)
IMAGE	ChatGPT or Gemini (try both!)
AUDIO	(We'll do this in Session 16!)

For TEXT: Try These Prompts

Use your planned prompt from Mini Activity 2, or try one of these:

What You Want	Prompt
Funny poem	"Write a funny birthday poem for my friend [friend's name] who loves [hobby/food]. Make it 8 lines with a funny ending and include rhymes."
Heartfelt message	"Write a heartfelt birthday message for my best friend. Mention that we've been friends for [X] years and why they're special to me. Make it warm and personal."
Invitation	"Create a birthday party invitation text for a [age]-year-old's party. Include: date (Saturday, March 15), time (2-5 PM), place (Community Center), RSVP info, and what to bring (swimsuits and towels). Make it exciting and fun!"
Joke collection	"Write 5 birthday-themed jokes that would make a 12-year-old laugh. Keep them clean and silly."
Short story	"Write a short story (about 150 words) about a birthday party where something magical happens. Include a cat, a cake, and a surprise."

For IMAGE: Try These Prompts

Use your planned prompt from Mini Activity 2, or try one of these:

What You Want	Prompt
Birthday card	"Create a colorful birthday card with a [dragon/unicorn/dinosaur/cat/dog] wearing a party hat and holding balloons. Include a birthday cake in the background with candles. Style: cute, colorful, digital art with bright colors and friendly look."
Coloring page	"Create a black and white line art coloring page for kids. Show a birthday party scene with a large cake, presents, balloons, and children playing games. No shading, just clear outlines."
Poster	"Design a birthday party poster for a [age]-year-old. Include space for date, time, and location text. Use bright colors, balloons, confetti, and party hats. Style: fun, celebratory, eye-catching."
Avatar	"Create a fun avatar of a [cat/dog/robot/animal] celebrating a birthday with a party hat and holding a small cake with one candle. Style: cartoon, cute, simple, perfect for a profile picture."
Digital gift tag	"Create a digital gift tag image with 'Happy Birthday' and [friend's name]. Include small illustrations of presents, balloons, and stars. Style: colorful, festive, printable."

Discussion: What Did You Create?

What did you create?

Describe your birthday artifact.

Which tool did you use?

ChatGPT, Gemini, or both?

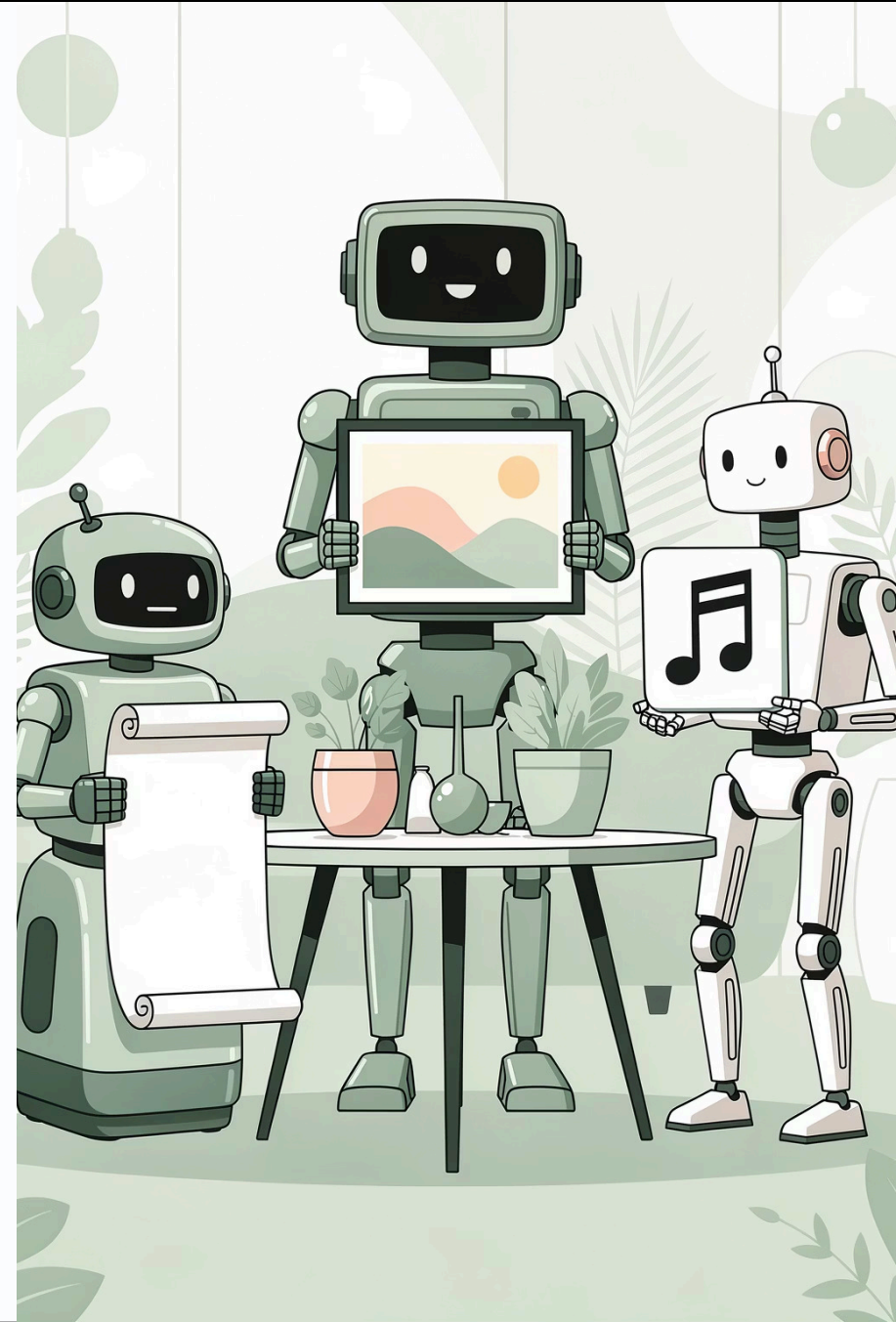
Did it turn out how you imagined?

What was similar? What was different?

What would you change?

Your prompt? The tool? The idea? How did your prompt from Mini Activity 2 work?

- ① AI is a creative partner - it takes your ideas and helps bring them to life. But YOU are the director! The quality of what you get depends on the quality of your prompt.



Summary : What We Learned Today

Concept	Key Idea
Generative AI	AI that CREATES new content (text, images, audio)
Traditional (Discriminative) AI	AI that ANALYZES or CLASSIFIES existing content
How it works	1. Trained on millions of examples → 2. Learns patterns → 3. Remixes patterns to create new things
Key insight	AI doesn't "understand" - it's a pattern remixer!
Types	Text, Image, Audio
Your role	You guide it with PROMPTS - your ideas shape what it creates

i Most Important Takeaway: You are the creator. AI is your tool. The more clearly you communicate your vision (through prompts), the better your results!

Homework : Part 1 - Quick Warm-Up

Match each term to its definition:

#	Term	Definition
1	Generative AI	A. AI that analyzes or classifies existing content
2	Discriminative AI	B. Your instruction to an AI about what to create
3	Prompt	C. AI that creates new content
4	Training Data	D. The examples AI learns from

Answers: 1-____, 2-____, 3-____, 4-____

Homework : Part 2 - Improve Your Birthday Artifact

Take your birthday artifact from today's activity and make it **BETTER!**

Review

Review what you created in class

Improve your prompt

Make it more specific, add details, clarify the style, include more about your friend

Try again

Try again with your improved prompt using the same tool

Compare

Compare the new version to the old one

	Original Prompt	Improved Prompt
Your Prompt		
What AI Created	(paste or describe)	(paste or describe)
What's better about the new version?		

Next Session Preview

Session 8: Large Language Models (LLMs)

You've used ChatGPT and Gemini to create birthday artifacts. Now let's understand what's inside these amazing tools!



What makes a "large" language model?

Hint: it's not just size!



How do they predict the next word?

Like super-powered autocomplete



What are parameters?

Billions of tiny dials inside



How were they trained?

Reading the entire internet!